

Yin Feng

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Beijing Jiaotong University, No.3 Shangyuancun, Haidian District, Beijing 100044, China

Education

Beijing Jiaotong University, China

Sept. 2025–Jun. 2028 (expected)

Transportation Planning and Management, School of Traffic and Transportation

M.S. Candidate

- First-semester GPA: 3.64/4.
- Core courses: Modern Freight Transportation Technology (100); Spatial Information Processing Methods and Technologies (93); Intelligent Transportation Systems Design and Integration (91); Urban Traffic Analysis and Control (90).

Chang'an University, China

Sept. 2021–Jun. 2025

Department of Transportation, School of Transportation Engineering

B.Eng. in Transportation

- GPA: 3.72/4.
- Core courses: Transportation Engineering Theory (96); Carriage of Dangerous Goods (95); Fundamentals of College Computer (94); GIS (Geographic Information System) (bilingual) (92); Traffic Safety Engineering (92); Engineering Mechanics (91); Automotive Operation Engineering (91); Operations Research (90).

Publications

1. Boyao Peng, Lexing Zhang, Enkai Li, **Yin Feng**, Ying Zhang, Li Zhong. Free-Flow Tolling System for Expressway with Fusion of 5G Communication and High-Precision Positioning Technology. *The 4th International Conference on Smart City Engineering and Public Transportation (SCEPT 2024)*. Accepted.
2. **Yin Feng**, Enjian Yao, Rui Zhang, Rongsheng Chen, Lexing Zhang. Optimization of Electric Vehicle Charging Station Layout Considering Mobile Charging Vehicles. *Transportation Research Part D: Transport and Environment*. Under minor revision.
3. Lexing Zhang, **Yin Feng**, Rongsheng Chen, Qi Ai, Michael Levin. Time-aware MPC Scheduling for Mobile Charging Vehicles with Fixed Charging Stations. *Transportation Research Part C: Emerging Technologies*. Under review.
4. Tian Xu, Lexing Zhang, Jiale Lei, **Yin Feng**, Wenxuan Wang, Yinen Ge. Trajectory-Level Acceleration Prediction at Tunnel Exits: A Dual-Input Seq2Seq Framework Incorporating Environmental and Individual Driving Behavior. *International Journal of Automotive Technology*. Under review.

Research Experience

Optimization of Electric Vehicle Charging Station Layout Considering Mobile Charging Vehicles

Project Leader | Xi'an, China

2025–Present

- Developed a welfare-oriented MILP model for hybrid EV charging infrastructure planning, jointly optimizing FCS locations, charging-pile capacities, MCV fleet size, and multi-period service allocation.
- Estimated spatially heterogeneous charging demand from urban POI data using kernel density estimation, and screened candidate FCS locations based on underserved-area analysis.
- Incorporated station capacity, MCV dispatch feasibility, inter-period inventory conservation, and minimum zone-level service requirements into the optimization model.
- Conducted a real district-level case study in Xi'an to compare fixed-only, mobile-only, and hybrid fixed-mobile schemes in terms of service coverage, unmet demand, user burden, and net social welfare.

Speed Optimization Control Strategy for Autonomous Vehicles in Highway Tunnel Areas Based on Hazard Perception

Project Leader | Xi'an, China

2024

- Developed a hazard-perception-based speed optimization strategy for autonomous vehicles in highway tunnel areas under mixed traffic conditions.
- Processed millimeter-wave radar trajectory data to analyze vehicle speed evolution, car-following behavior, and potential rear-end collision risks near tunnel entrances and exits.
- Designed a Seq2Seq neural-network model to predict short-term vehicle speeds and combined it with lane-change timing prediction for proactive speed-control decision-making.
- Built a simulation-based validation framework to evaluate safety improvements in tunnel transition areas.

Competitions

- Meritorious Winner, Mathematical Contest in Modeling (MCM) Mar. 2024
- Provincial 1st Prize, The 1st Shaanxi Provincial College Students Transportation Technology Competition Apr. 2023
- Provincial 2nd Prize, The 1st Shaanxi Provincial College Students Transportation Technology Competition Apr. 2024
- Provincial 3rd Prize, The 3rd Shaanxi Provincial College Students Transportation Technology Competition Apr. 2024

- University-Level Gold Award, The 9th China International College Students' "Internet+" Innovation and Entrepreneurship Competition Sept. 2023

Scholarships

- First-Class Scholarship, 2025–2026 academic year.
- National Encouragement Scholarship, 2023–2024 academic year.
- National Encouragement Scholarship, 2021–2022 academic year.
- College Contribution Scholarship, 2022–2023 academic year.
- Course Excellence Scholarship, 2022–2023 academic year.

Skills

- Software: SPSS, Visio, VISSIM, AutoCAD, Origin.
- Computer skills: Python; familiar with basic machine learning algorithms and data processing.
- Languages: CET-4 468; CET-6 455.